

Research Proposal

**Enhancing the competency of using
problem solving strategy in mathematics
among senior secondary teachers in Sri Lanka**

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Research Proposal

Research Topic

**Enhancing the competency of using
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among senior secondary teachers in Sri Lanka**

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Enhancing the Competency of using problem solving strategy in mathematics among senior secondary teachers in Sri Lanka

1 Introduction

1.1 Background of the study

Mathematics education plays a pivotal role in shaping the analytical and critical thinking skills of students, which are essential for their success in an increasingly complex world. In Sri Lanka, the quality of mathematics instruction at the senior secondary level is crucial, as it lays the foundation for students' future academic and professional pursuits. However, many educators face challenges in effectively teaching mathematics, particularly in employing problem-solving strategies that foster deeper understanding and engagement among students.

Problem-solving is not merely a skill but a fundamental aspect of mathematical learning that encourages students to explore, conjecture, and reason. It empowers them to approach mathematical concepts with confidence and creativity. Despite its importance, research indicates that many senior secondary teachers in Sri Lanka may lack the necessary competencies to effectively implement problem-solving strategies in their classrooms. This gap can lead to a less engaging learning environment and hinder students' ability to develop critical mathematical skills.

This research aims to enhance the competency of senior secondary mathematics teachers in Sri Lanka by focusing on effective problem-solving strategies. Through this research, we hope to illuminate the pathways for enhancing pedagogical practices and nurturing a generation of students who are proficient in mathematics and prepared for the challenges of the future.

Mathematics teaching provides students with the ability of critical thinking and problem solving. The effective teaching of mathematics is critical for developing students' problem-solving skills, which are essential for their academic success and future careers. The implementation of problem-solving strategies in mathematics classrooms can be significantly influenced by teachers. Many teachers face challenges in employing problem-solving strategies that foster deeper understanding and engagement among students.

Teachers have to upgrade their knowledge in enhancing problem-solving strategies in secondary mathematics teaching in Sri Lanka to meet the challenges in the modern mathematical world.

The latest international studies and some cognitive research studies suggest that there is a space for the improvement in the mathematics knowledge of the teachers (ICMI, 2016). A number of studies report that teachers' mathematics content knowledge is insufficient, leaving them underprepared to teach in ways that help students internalize salient concepts (Siregar, 2017). In Sri Lanka, the National Education Commission (2018) report says that teachers who are appointed to teach mathematics for students fail to upgrade their knowledge. Department of Examination (2019) report also states that the performance of teachers in mathematics is poor.

1.2 Research Problem

In Sri Lanka, senior secondary mathematics teachers face significant challenges particularly in the effective implementation of problem-solving strategies. Although many teachers possess subject knowledge in mathematics they struggle to integrate problem-solving techniques into their teaching practices. As such a lack of student engagement and poor performance in mathematics is evident. This gap in pedagogical skills not only affects students' critical thinking abilities but also hinders the ability of teachers to effectively teach and impart mathematical skills to students. Thus, there is an urgent need to enhance the competency of senior secondary teachers in using problem-solving strategies to foster a more interactive and effective learning environment, ultimately improving student outcomes in mathematics across the region. This study aims to investigate and enhance the competency of senior secondary teachers in using problem-solving strategies in mathematics education.

2 Literature Review

- ◆ **Theoretical Framework:** This section will review the relevant theories on attitudes and beliefs, including the Theory of Planned Behavior and Self-Efficacy Theory.
- ◆ **Previous Studies:** A review of previous research on the impact of teachers' attitudes and beliefs on instructional practices, specifically in mathematics education.
- ◆ **Problem-Solving Strategies in Mathematics:** An overview of effective problem-solving strategies and their importance in secondary education.
- ◆ **Contextual Factors in Sri Lanka:** A discussion on the educational context in Sri Lanka, including cultural, societal, and institutional influences on teaching practices.

The study focuses its attention mainly on the above topics. The following table displays the relevant studies under these topics

Author	Study title	Finding	Relevant & Current study
NIE, 2019	Teacher' mathematics knowledge	Many teachers are without required knowledge and trainings to teach mathematics	It is essential to ensure all mathematics teachers are subject-competent and have good pedagogical knowledge to use problem solving strategies to teach mathematics effectively.
DOE, 2018	Teachers' pedagogical knowledge	Many teachers fail to use proper pedagogy while teaching mathematics.	
Vygotsky, 1978	Collaborative Learning	Studies have shown that collaborative approaches, where students work in groups to solve problems, enhance problem-solving skills. Vygotsky's Social Development Theory emphasizes the role of social interaction in cognitive development, supporting this finding	Design and implement an intervention to enhance teachers' competency in applying problem-solving strategies in mathematics
Lesh et al., 2019	Problem-Solving Strategies in Mathematics	An overview of effective problem-solving strategies and their importance in secondary education.	Empower mathematics problem solving competency in Sri Lanka

Table 1: Relevant studies under the review topics

3 Research Objectives

- I. Identify the current level of cognitive abilities of secondary mathematics teachers regarding problem-solving strategies.
- II. Explore the level of use of problem solving strategies in mathematics by senior secondary teachers.
- III. Find out areas for improvement in problem-solving strategy among senior secondary teachers.
- IV. Design and implement an intervention to enhance teachers' competency in applying problem-solving strategies in mathematics.
- V. Assess the progress of use of problem-solving strategies in mathematics after intervention.
- VI. Propose a teacher education module on effective use of problem solving strategies in mathematics in senior secondary education.

4 Methodology

Research design

Multi-phase mixed-methods research design

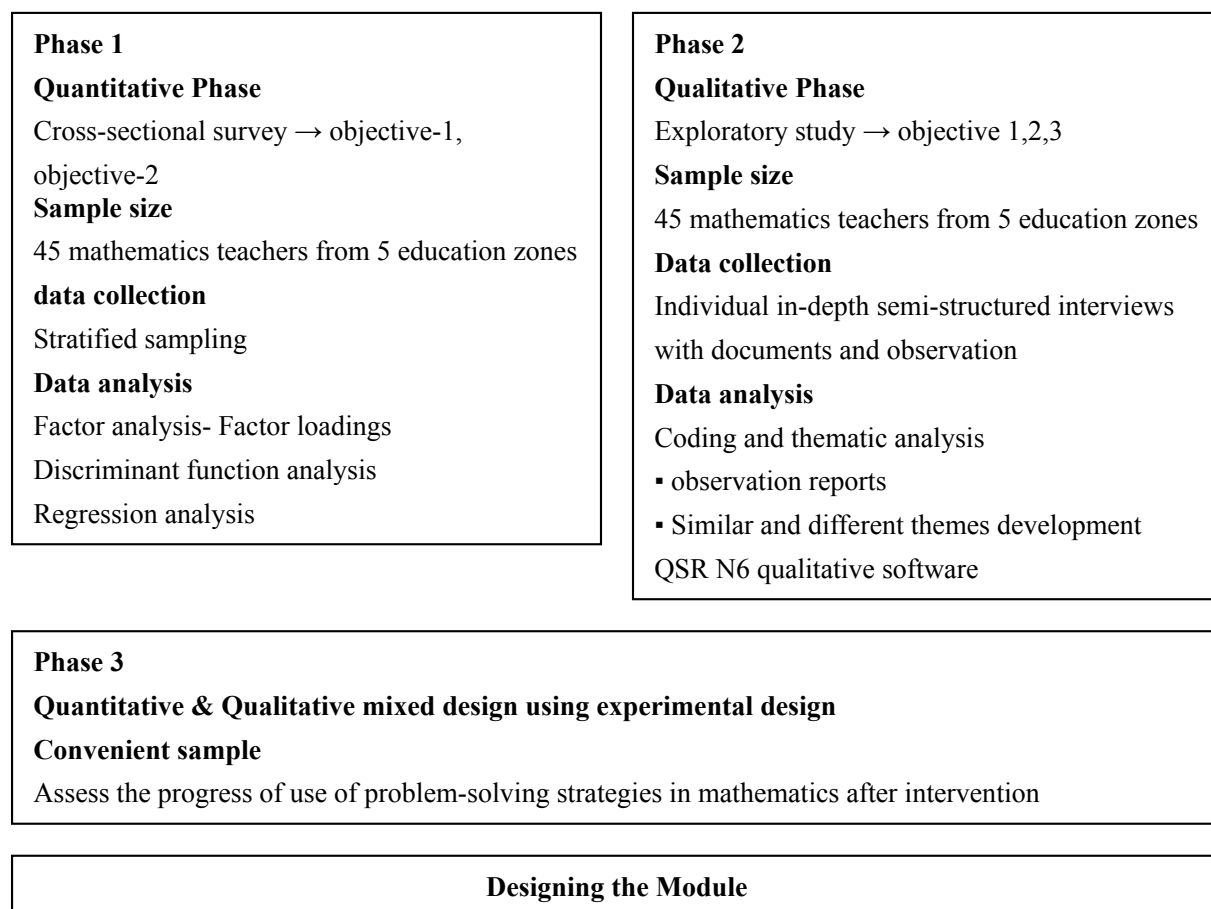


Figure 1: Multi-phase mixed-methods research design

4.1 Research Design

This study will employ a multi-phase mixed-methods research design, combining quantitative surveys and qualitative data to provide a comprehensive understanding of the research questions. The study is structured through three phases. To conduct the above study through a multi-phase mixed methods design with an action study as a component in one phase following these steps:

1. Phase 1: Quantitative Survey

1.1 Design: Develop a quantitative survey questionnaire to assess teachers' attitudes towards mathematics and their implementation of problem-solving strategies. Include items that measure perceptions of problem-solving strategies, perceived effectiveness, barriers, and facilitators.

1.2 Sampling: Use purposive sampling to select a diverse sample of mathematics teachers who teach senior secondary mathematics. Aim to recruit a sufficient number of participants to ensure statistical power.

1.3 Data Collection: Administer the survey questionnaire to the selected participants either online or in person, depending on logistical considerations. Ensure that participants have adequate time to complete the survey and emphasize the importance of honest and thoughtful responses.

1.4 Data Analysis: Analyze the quantitative data using descriptive statistics to identify patterns and trends in teachers' attitudes towards mathematics and their implementation of problem-solving strategies. Explore differences in attitudes based on demographic variables such as teaching experience, educational background, and institutional context.

2. Phase 2: Qualitative Interviews

2.1 Design: Develop a semi-structured interview protocol to explore teachers' experiences, perceptions, and attitudes towards mathematics and their implementation of problem-solving strategies in more depth. Include open-ended questions that allow participants to elaborate on their views and provide examples from their teaching practice.

2.2 Sampling: Select a subset of participants from the survey sample to participate in qualitative interviews. Aim for diversity in terms of attitudes towards mathematics and their implementation of problem-solving strategies and teaching contexts.

2.3 Data Collection: Conduct individual or group interviews with the selected participants to explore their perspectives on self-learning and its relevance to preparing to teach mathematics in G.C.E advanced level classes. Record the interviews and take detailed notes to ensure accuracy in data collection.

2.4 Data Analysis: Analyze the qualitative data using thematic analysis to identify recurring themes and patterns in teachers' narratives. Compare and contrast the qualitative findings with the quantitative results to gain a comprehensive understanding of teacher attitudes towards mathematics and their implementation of problem-solving strategies.

3. Phase 3: Quantitative & Qualitative mixed design using experimental design Convenient sample

Assess the progress of use of problem-solving strategies in mathematics after intervention.

3.1 Design: Integrate an experimental research component into the study to actively engage teachers in the process of attitudes towards mathematics and their implementation of problem-solving strategies and professional development. Collaborate with participating teachers to design and implement self-learning interventions aimed at enhancing their preparation for teaching mathematics in senior secondary classes.

3.2 Implementation: Work closely with the participating teachers to implement their problem-solving strategies interventions over a specified period. Provide support, resources, and guidance as needed to facilitate their engagement in self-directed learning activities.

3.3 Data Collection: Gather data on the implementation of the problems solving interventions, including teachers' experiences, challenges encountered, and perceived impact on their teaching practice. Use a combination of observation, reflection, and documentation to capture relevant information.

3.4 Evaluation: Evaluate the effectiveness of the attitudes towards mathematics and their implementation of problem-solving strategies interventions in enhancing teachers' preparation for teaching mathematics. Collect feedback from participating teachers and stakeholders to inform future iterations of the interventions.

Participants

- **Population:** Secondary school mathematics teachers in Sri Lanka.
- **Sample:** A Convenient sample of 150 teachers from various regions, ensuring representation from urban, suburban, and rural schools.

4.2 Data Collection

1. **Survey:** A structured questionnaire will be administered to collect quantitative data on teachers' attitudes, beliefs, and reported practices.
2. **Interviews:** Semi-structured interviews with a subset of 30 teachers to gain in-depth qualitative insights into their experiences and challenges.
3. **Classroom Observations:** Observations of 20 mathematics classes to examine the practical implementation of problem-solving strategies and triangulate data from surveys and interviews.

4.3 Data Analysis

1. **Quantitative Data:** Statistical analysis (e.g., correlation, regression) to identify relationships between teachers' attitudes, beliefs, and their instructional practices.
2. **Qualitative Data:** Thematic analysis of interview transcripts and observation notes to identify common themes and patterns.

5 Potential contributions and implications

Teachers with good mathematics knowledge and effective teaching practices are needed for imparting excellent mathematics education. This study contributes to the development of effective professional development programmes for mathematics teachers and inform policy decisions in mathematics education through the keen identification of key challenges and barriers to implementing problem-solving strategies. This study also states recommendations for professional development programmes to support teachers in overcoming these challenges. This study contributes to design and implement an intervention to empower problem solving competency and enhance teacher's competency in applying problem-solving strategies in mathematics. It contributes to students' academic success by proposing a teacher education module.

6 Research Timeline

Jan 2025 – May 2026

Work	J	F	M	A	M	J	JL	A	S	O	N	D	J	F	M	A	M
Reading literature reviews																	
Writing 1 st and 2 nd Chapters																	
Developing data collection tools																	
Data collection																	
Data processing																	
Data analysis																	
Report writing																	
Thesis submission																	

Table 2: Research timeline (Gantt chart)

7 Conclusion

By identifying existing challenges and providing targeted professional development, this study seeks to equip educators with the tools and methodologies necessary to foster a more interactive and problem solving approach to mathematics instruction. Ultimately, the goal is to improve not only teacher effectiveness but also student outcomes in mathematics, thereby contributing to the overall advancement of education in Sri Lanka.

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